

Homework 3

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1. Let $\mathbb{P}^2$ be the projective plane. A line is defined to be the zero set of

$$\alpha X + \beta Y + \gamma Z = 0.$$

- (a) Show directly from the definition that two distinct points in $\mathbb{P}^2$ are contained in a unique line.
- (b) Similarly, show that any two distinct lines in $\mathbb{P}^2$ intersect in a unique point.

Proof.

- (a) Let $\langle x_1 : y_1 : z_1 \rangle$ and $\langle x_2 : y_2 : z_2 \rangle$ be distinct points. Then since (x_1, y_1, z_1) is not a scalar multiple of (x_2, y_2, z_2) , they are linearly independent over the reals. Any line that goes through them satisfies the equation

$$\begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and thus the null space of the linear transformation represented by the matrix on the left is 1-dimensional, meaning it is a unique line.

- (b) Say we have two lines

$$L_1 : \alpha_1 X + \beta_1 Y + \gamma_1 Z = 0$$

$$L_2 : \alpha_2 X + \beta_2 Y + \gamma_2 Z = 0.$$

Then we can break each one up into two cases which we can represent nicely in projective coordinates: one where $Z = 0$ and one where $Z \neq 0$. If $Z \neq 0$, we can divide the equations by Z and rename X and Y to get

$$L_1 : \alpha_1 X + \beta_1 Y + \gamma_1 = 0$$

$$L_2 : \alpha_2 X + \beta_2 Y + \gamma_2 = 0.$$

If $Z = 0$, we get

$$L_1 : \alpha_1 X + \beta_1 Y = 0$$

$$L_2 : \alpha_2 X + \beta_2 Y = 0.$$

Assume $\beta_1 \neq 0$ and $\beta_2 \neq 0$. Then we can write our lines in projective coordinates as

$$\begin{aligned} L_1 &: \left\{ \left\langle X : -\frac{\alpha_1}{\beta_1}X - \frac{\gamma_1}{\beta_1} : 1 \right\rangle, \left\langle X : -\frac{\alpha_1}{\beta_1}X : 0 \right\rangle \right\} \\ L_2 &: \left\{ \left\langle X : -\frac{\alpha_2}{\beta_2}X - \frac{\gamma_2}{\beta_2} : 1 \right\rangle, \left\langle X : -\frac{\alpha_2}{\beta_2}X : 0 \right\rangle \right\}. \end{aligned}$$

For the points at infinity, equality seems to come only when $\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$, but in that case, the equality of affine points would imply that $\frac{\gamma_1}{\beta_1} = \frac{\gamma_2}{\beta_2}$, which implies that the two equations are linear multiples of each other, contradicting the fact that they're distinct lines. Taking the left equality then, we deduce that

$$X = \frac{\frac{\gamma_1}{\beta_1} - \frac{\gamma_2}{\beta_2}}{\frac{\alpha_2}{\beta_2} - \frac{\alpha_1}{\beta_1}},$$

which determines a single point of intersection.

I don't have time to work out the case(s) where $\beta_i = 0$ for some i .

□

2. For each of the pairs below, find all of the points of intersection between the curves C_1 and C_2 in $\mathbb{P}^2(\mathbb{C})$. (So include complex points and points at infinity.)

- (a) $C_1 : x - y = 0, C_2 : x^2 - y = 0$.
- (b) $C_1 : x - y - 1 = 0, C_2 : x^2 - y^2 + 2 = 0$.
- (c) $C_1 : x - y - 1 = 0, C_2 : x^2 - 2y^2 - 5 = 0$.
- (d) $C_1 : x - 2 = 0, C_2 : y^2 - x^3 + 2x = 0$.

Proof.

- (a) The points on C_1 can be written as $\langle x : x : 1 \rangle$ or $\langle 1 : 1 : 0 \rangle$. We need to homogenize C_2 as $x^2 - yz = 0$, so the points on C_2 are of the form $\langle x : x^2 : 1 \rangle$ or $\langle 0 : 1 : 0 \rangle$. They share no points at infinity, then, and we are reduced to the case where $x = x^2$, so $x = 0$ or $x = 1$. Thus, our points of intersection are $\langle 0 : 0 : 1 \rangle$ and $\langle 1 : 1 : 1 \rangle$.
- (b) First, we need to homogenize both equations:

$$\begin{aligned} C_1 &: x - y - z = 0 \\ C_2 &: x^2 - y^2 + 2z^2 = 0. \end{aligned}$$

We then describe the points on C_1 and C_2 in projective coordinates:

$$\begin{aligned} C_1 &: \{ \langle x : x - 1 : 1 \rangle, \langle 1 : 1 : 0 \rangle \}, \\ C_2 &: \{ \langle x : \pm\sqrt{x^2 + 2} : 1 \rangle, \langle 1 : 1 : 0 \rangle \}. \end{aligned}$$

They agree at infinity, then. Back in the affine plane, we solve

$$\begin{aligned}x - 1 &= \pm\sqrt{x^2 + 2} \\x^2 - 2x + 1 &= x^2 + 2 \\2x &= -1 \\x &= -\frac{1}{2},\end{aligned}$$

but plugging $-\frac{1}{2}$ back into x , we get

$$\pm\sqrt{\left(-\frac{1}{2}\right)^2 + 2} = \pm\sqrt{\frac{9}{4}} = \pm\frac{3}{2}$$

and

$$-\frac{1}{2} - 1 = -\frac{3}{2},$$

so we get agreement only when taking the negative square root. Finally, we can characterize all points of intersection as

$$\left\{ \left\langle -\frac{1}{2} : -\frac{3}{2} : 1 \right\rangle, \langle 1 : 1 : 0 \rangle \right\}.$$

(c) First, homogenize both equations:

$$\begin{aligned}C_1 : x - y - z &= 0 \\C_2 : x^2 - 2y^2 - 5z^2 &= 0.\end{aligned}$$

Now, express them in coordinate form:

$$\begin{aligned}C_1 : & \{ \langle x : x - 1 : 1 \rangle, \langle x : x : 0 \rangle \} \\C_2 : & \left\{ \left\langle x : \pm\sqrt{\frac{x^2 - 5}{2}} : 1 \right\rangle, \left\langle x : \frac{x}{\sqrt{2}} : 0 \right\rangle \right\}.\end{aligned}$$

These agree on no points at infinity, but in the affine plane we have

$$\begin{aligned}x - 1 &= \sqrt{\frac{x^2 - 5}{2}} \\ \implies x^2 - 2x + 1 &= \frac{x^2 - 5}{2} \\ \implies 2x^2 - 4x + 2 &= x^2 - 5 \\ \implies x^2 - 4x + 7 &= 0 \\ \implies (x - 2)^2 + 3 &= 0 \\ \implies x &= \pm\sqrt{3}i + 2.\end{aligned}$$

To see if both solutions work, evaluate

$$\begin{aligned}
\sqrt{\frac{(\pm\sqrt{3}i + 2)^2 - 5}{2}} &= \sqrt{\frac{\pm 4\sqrt{3}i - 4}{2}} \\
&= \sqrt{\pm 2\sqrt{3}i - 2} \\
&= \sqrt{-3 \pm 2\sqrt{3}i + 1} \\
&= \sqrt{(\pm\sqrt{3}i + 1)^2} \\
&= \pm\sqrt{3}i + 1 \\
&= (\pm\sqrt{3}i + 2) - 1.
\end{aligned}$$

Thus, our two points of intersection are $\langle \pm\sqrt{3}i + 2 : \pm\sqrt{3}i + 1 : 1 \rangle$.

(d) First, homogenize:

$$\begin{aligned}
C_1 : x - 2z &= 0 \\
C_2 : y^2z - x^3 + 2xz^2 &= 0.
\end{aligned}$$

Then, write them in coordinate form:

$$\begin{aligned}
C_1 : \{ \langle 2 : y : 1 \rangle, \langle 0 : 1 : 0 \rangle \} \\
C_2 : \{ \langle x : \pm\sqrt{x^3 - 2x} : 1 \rangle, \langle 0 : 1 : 0 \rangle \}.
\end{aligned}$$

These clearly agree on their point at infinity, and the only other possible point would be when $x = 2$. The y -coordinate is free in C_1 , so we calculate that the y -coordinate of C_2 is

$$\pm\sqrt{2^3 - 2 \cdot 2} = \pm\sqrt{8 - 4} = \pm\sqrt{4} = \pm 2.$$

Thus, our three points of intersection are $\langle 2 : 2 : 1 \rangle$, $\langle 2 : -2 : 1 \rangle$, and $\langle 0 : 1 : 0 \rangle$.

□

3. Find a parametrization for all rational points on the curve $y^2 = x^3 - 2x^2 - 4x + 8$. (Hint: There is a point on the curve, a singular point, with the property that any line through that point intersects the curve with multiplicity 2. Because this curve is singular, it has genus 0.)

Proof. We can factor the equation as

$$y^2 = (x - 2)^2(x + 2).$$

This tells us two important things. First, $(2, 0)$ is our singular point. Second, if (x, y) is a rational point on the curve, then $y = \pm(x - 2)\sqrt{x + 2}$ is rational, so in particular $\sqrt{x + 2}$ is rational.

Now take the general form of a line through $(2, 0)$, namely $y = m(x - 2)$, where m is our slope. Intersecting this line with the curve, we get

$$m^2(x - 2)^2 = (x - 2)^2(x + 2),$$

implying $x = m^2 - 2$ whenever $x \neq 2$. Then

$$y = m(m^2 - 4) = m^3 - 4m,$$

so we can parametrize the rational points other than $(2, 0)$ as

$$\langle m^2 - 2, m^3 - 4m \rangle, m \in \mathbb{Q}.$$

Certainly, all points parametrized in this way are rational, but how do we know this parametrizes *all* rational points on the curve? Take a general rational point (x, y) and $m = \pm\sqrt{x+2}$. Then the resulting point in the parametrization is

$$\begin{aligned} \langle (x+2) - 2, \pm\sqrt{x+2}^3 - 4\pm\sqrt{x+2} \rangle &= \langle x, \pm(\sqrt{x+2})(x+2-4) \rangle \\ &= \langle x, \pm(\sqrt{x+2})(x-2) \rangle \\ &= \langle x, y \rangle, \end{aligned}$$

so we have produced a rational slope which parametrizes an arbitrary rational point, and thus we have parametrized all rational points on the curve. $\square$